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Horizontal Well Design Optimization in Tight Reservoirs

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Abstract

Tight reservoirs pose significant challenges to drilling and completions and are often developed with horizontal wells and hydraulic fracturing stimulations. Designing well trajectory plans for horizontal wells needs to consider the risks of low rate of penetration (ROP) and wellbore instability while drilling, hydraulic fracture geometry and fracture conductivity while designing completions, and the risks of solids production while producing hydrocarbon from the wells. This paper aims to provide an integrated horizontal well orientation design method with various considerations during drilling, completion, and production stages.

ROP is influenced by various formation properties, drilling conditions, and bit parameters. Predicting ROP within a 3D model offers a valuable opportunity to place wells in the most drillable rock. We will explain our method of 3D ROP prediction and how the predicted ROP can be used for optimizing well trajectory design. Based on analytical equations, we explain how the well's azimuth angles affect the risks of wellbore instability in both drilling and production stages of horizontal wells and the criteria for determining the most stable wellbore and perforation orientations in different stress regimes. Hydraulic fracture geometry and conductivity are related to wellbore trajectory, in-situ stresses, rock fabric, and mechanical properties of horizontal wells. Optimized well trajectory can improve hydraulic fracture geometry and dimensionless fracture conductivity while reducing near-wellbore tortuosity. Wellbore trajectory requirements for achieving optimum hydraulic fracture geometry and stimulation results will be explained. A case study is provided to illustrate the integrated process of well trajectory optimization considering ROP, wellbore instability, solids production, and hydraulic fracture geometry.

In the case study, two horizontal wells were planned in the same reservoir zone. In-situ stress data was used to determine the well orientation in which the wellbores are most stable while drilling and the perforations are most stable while producing. The 3D ROP model was used to optimize drilling performance. The final optimum well trajectories were determined considering wellbore and perforation stability, drilling performance, and hydraulic fracture geometry. The drilling results validated the integrated approach for selecting the optimum horizontal well orientation.

Introduction

Deep and tight reservoirs have been extensively developed in various parts of the world. These reservoirs are characterized by low porosity and permeability, often bearing light oil, condensate, or dry gas. They

are typically found in tight sandstone, carbonate rocks, or unconventional formations such as shale. Tight reservoirs present various challenges in different phases of reservoir development. These include a low rate of penetration (Najwani et al., 2013) and wellbore instability (Li et al., 2012) in drilling, inefficient hydraulic fracturing in completing the wells (Soliman et al., 1990), and solids production during production (Morita and Boyd, 1991; Abass et al., 2003).

Tight reservoirs are typically composed of formation rocks with high rock strength and rocks with unconfined compressive strength (UCS) values of 35,000 psi or higher have been reported (Najwani et al., 2013). These rocks present unique challenges of low ROP while drilling. ROP as low as 1 ft/hr is common in drilling these formations. High quartz content in tight clastic reservoirs combined with high rock strength make these reservoir rocks very abrasive and very low footage can be drilled with any type of drill bit in these formations before the bit is worn out.

Wellbore instability often occurs as a result of wellbore failure due to stress concentrations at the wellbore wall exceeding rock strength. There is higher propensity for wellbores to fail in rocks with higher porosity and lower rock strength, such as unconsolidated or weakly consolidated formations or shale formations. However, even rock strength in deep and tight formations is generally high, because of the higher in-situ stresses in those formations, wellbore failure also frequently occurs (Li et al., 2012). The risk of wellbore instability is different for horizontal wellbores in different directions, depending on the stress regime, orientation, and the magnitudes of the in-situ stresses.

Solids production is a result of wellbore or perforation tunnel failure while producing from a well. When a well is producing, there is a drawdown pressure between the reservoir and the wellbore, so that the formation fluid can flow into the well. Contrary to the overbalanced drilling condition where there is a mud cake to isolate the reservoir pressure from the wellbore pressure, the reservoir pressure at the wellbore wall is equal to the well pressure in a production well. Because of the lower well pressure compared to the drilling conditions and the lack of a differential pressure to support the borehole wall, the effective hoop stress concentrated at the wellbore wall can become greater while the effective rock strength becomes smaller. The more excessive stress concentration at the wellbore wall and the lower rock strength make a wellbore more prone to wellbore failure while producing, especially when the reservoir pressure is depleted. In a cased and perforated wellbore, the perforation tunnels are under the in-situ stresses, and stress concentration is generated around the perforation tunnels. Perforations also fail when such stress concentration exceeds rock strength. When a wellbore or the perforations fail, the failed rock material enters the wellbore. When fluid flow velocity is adequate to transport the loose rock material to the surface, solids production is observed from the surface. Solids production is commonly encountered in unconsolidated or weakly consolidated rocks, while it can also occur in tight formations due to higher stresses (Morita and Boyd, 1991). The optimum horizontal wellbore or perforation direction depends on the stress regime and the magnitudes of the in-situ stresses.

Hydraulic fracturing stimulation is frequently used in tight reservoirs to stimulate the reservoir so that an economic production rate can be achieved in otherwise uneconomic reservoirs. Hydraulic fracture geometries in horizontal wells can be very complex and such complexity is related to the horizontal well orientation in relation to the in-situ stress direction, the orientation of the perforation holes in cased wellbores, the magnitudes of the in-situ stresses and rock textures (Abass et al., 1996; Soliman et al., 1990; Yew and Weng, 2015). Aligning the wellbore direction with horizontal stress directions is typically critical for achieving optimum hydraulic fracture geometry. When the wellbore is aligned with the maximum horizontal stress direction, longitudinal fractures are created and propagate along the direction of the wellbore axis. These longitudinal fractures are typically applied in higher porosity and permeability rocks. These fractures are mainly for bypassing near-wellbore damage zones instead of creating large fracture surface areas. However, in tight reservoirs where the formation permeability is typically low, increasing fracture surface area is more critical, and the wellbores are more commonly aligned with the minimum horizontal stress direction so that multiple transverse fractures are created. Horizontal wellbores not aligned

with either horizontal stress direction tend to create ineffective hydraulic fractures as fracture realignment with stresses in the near-wellbore region creates near-wellbore tortuosity, which could limit the propagation of hydraulic fractures and the transport of proppant.

Horizontal Well Design for ROP Optimization

Low ROP is a common challenge in drilling tight reservoirs. Even though ROP is generally low in tight formations, due to reservoir heterogeneity, there are zones and regions in a reservoir that are relatively easier to drill than others. A unique approach of predicting the 3D distribution of ROP based on the mechanical specific energy concept considering the effects of rock strength and other properties, abrasiveness, pore pressure and the effects of drilling parameters and bit selection was developed and has been successfully applied in a tight reservoir for optimizing ROP (Li et al., 2024). The method integrates multiple data sources, including rock mechanical and petrophysical properties, drill bit data, drilling fluid and drilling mechanics data, and predicts ROP distribution in a 3D volume based on the concept of mechanical specific energy and hydraulic energy. The predicted ROP considers the effects of rock strength, rock abrasiveness, pore pressure, drilling, and bit parameters. The distribution of ROP in a reservoir helps to plan horizontal wells in the zones and regions of reservoir rocks with higher predicted ROP for achieving better drilling performance.

The method of 3D ROP prediction starts with establishing a relationship between formation abrasiveness and formation properties and drilling parameters based on actual well data. The formation abrasiveness is further used to predict bit wear, which is subsequently used in predicting the ROP, so that the bit wear effect is considered in the predicted ROP. Eq. (1) shows the relationship between the drill bit dull grade, representing bit wear, with the formation and drilling parameters.

$$D = f W_b a N^b T^c V_q d \xi_c^e \phi_i^g S^h H^i \quad (1)$$

where W_b is downhole weight on bit, klb; N is rotary speed of the drill bit, revolutions per minute; H is number of running hours of the drill bit, hour; T is a parameter representing the relative rock grain size, dimensionless; V_q is volume of quartz, dimensionless; ξ_c is confined compressive strength, psi; ϕ_i is porosity, dimensionless; S is water saturation, dimensionless; D is drill bit dull grade; a , b , c , d , e , g , h , i , and f are correlation coefficients.

The correlation coefficients from Eq. (1) can be derived using multivariate linear regression using actual well data from multiple bit runs from a selected number of wells drilled in the same reservoir.

Eq. (2) shows the ROP prediction equation based on mechanical and hydraulic specific energy using the input data from drilling, drilling fluid, drill bit and formation properties.

$$ROP = \frac{\frac{13.33 \mu_b N \times W e^{-\mu \beta}}{d_b} \left(1 + \frac{\frac{1154 \Gamma \Delta P_b Q}{A_b}}{\frac{13.33 \mu_b N \times W e^{-\mu \beta}}{d_b}} \right)}{\frac{\xi_c}{E_m} - \frac{W e^{-\mu \beta}}{A_b}} \left(1 - \frac{\Delta d \times D}{8d} \right) \quad (2)$$

where W is surface weight on bit, lbf; ΔP_b is pressure drop across the bit, psi; Q is flow rate, gallon per minute; μ_b is bit-specific coefficient of sliding friction, dimensionless; Γ is hydraulic energy reduction factor, dimensionless; d_b is bit diameter, inches; A_b is bit area, square inches; μ is friction coefficient between drill pipe and wellbore wall, dimensionless; E_m is mechanical efficiency; β is well inclination angle, degree; Δd is footage drilled by a drill bit, ft and d is total footage drilled by a drill bit, ft.

Mechanical efficiency is a critical parameter in the prediction of ROP, and it is estimated by matching predicted ROP with actual ROP in existing wells. Implementing the formation abrasiveness and ROP prediction method in a 3D model populated with required petrophysical properties, mechanical properties, pore pressure, and other required data, the 3D distribution of abrasiveness and ROP can be predicted. Once

the 3D ROP and abrasiveness distribution are available, they can be used for planning horizontal wells and optimizing drilling performance. The method of using the ROP prediction for planning horizontal wells is illustrated in Li et al. 2024. Fig. 1 shows an example of the predicted 3D distribution of formation abrasiveness and ROP in a tight clastic reservoir in the Middle East.

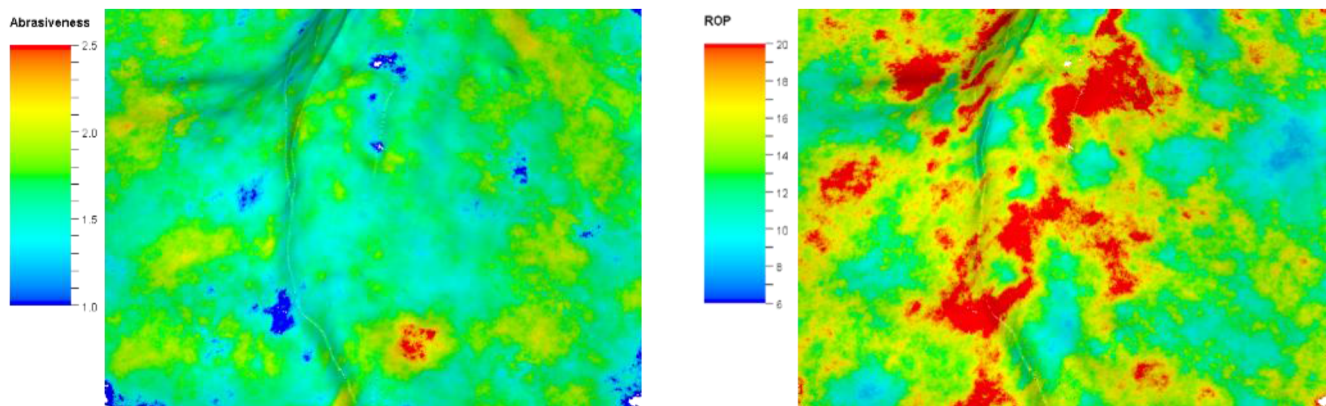


Figure 1—Example of predicted formation abrasiveness (left) and ROP (right) distribution in a 3D model. The formation abrasiveness values are relative, and low values indicate low formation abrasiveness. The predicted ROP values already considered the predicted formation abrasiveness, and low values indicate low expected ROP.

Horizontal Well Design for Minimizing Wellbore Instability

Wellbore stability while drilling is a function of the stress concentration at the wellbore wall, well pressure, pore pressure, and rock strength. Assuming that there is perfect mud cake on the wellbore wall and therefore no pressure communication between the wellbore and the formation, in the cross-section of a wellbore, the maximum stress concentration is found in the direction of the minimum far-field stress. The maximum stress concentration can be calculated as:

$$\sigma_{\theta} = 3\sigma_1 - \sigma_2 - P_w \quad (3)$$

where σ_{θ} is the maximum hoop stress at the wellbore wall, σ_1 is the maximum far-field stress and σ_2 is the minimum far-field stress in the cross-section plane of the wellbore, and P_w is well pressure.

In a horizontal well, one of the two far-field stresses in Eq. (3) is the vertical stress, and the other is the horizontal stress. Depending on the azimuth of the wellbore, the horizontal stress in the wellbore cross-section plane varies between the minimum horizontal stress and the maximum horizontal stress. The optimum horizontal wellbore direction for wellbore stability is the direction in which the hoop stress is the least. We need to know the stress regime before determining the optimum horizontal well direction. In a normal fault stress regime where vertical stress is the maximum in-situ stress, the optimum wellbore orientation is the minimum horizontal stress direction. In a reverse fault stress regime where vertical stress is the minimum far-field stress, the optimum wellbore orientation is the maximum horizontal stress direction. In a strike-slip stress regime, vertical stress is the intermediate far-field stress, and the optimum wellbore direction depends on the magnitudes of the far-field stresses. Using Eq. (3) and considering the different combinations of maximum and minimum far-field stresses in the wellbore cross-section plane when the wellbore is oriented in both maximum horizontal stress direction and minimum horizontal stress direction, respectively, we can determine that when Eq. (4) is satisfied, the most stable wellbore direction is the maximum horizontal stress direction and otherwise the most stable wellbore direction is the minimum horizontal stress direction.

$$\sigma_H > \frac{1}{3}(4\sigma_V - \sigma_h) \quad (4)$$

where σ_H is the maximum horizontal stress, σ_h is the minimum horizontal stress, and σ_V is the vertical stress.

If the mud cake is imperfect and there is fluid flow across the wellbore wall, the hoop stress calculation takes a different form than Eq. (3) (Li, 2023). However, the optimum wellbore orientations in different stress regimes remain the same.

Horizontal Well Design for Minimizing Solids Production

Solids production while producing hydrocarbon from a wellbore is a result of wellbore failure in an open hole completion or perforation failure in a cased and perforated completion. When fluid velocity in the wellbore is high enough to bring the failed rock material out of the well, solids production is observed at the surface. When reservoir fluid is being produced from a wellbore, there is fluid flow and pressure communication across the wellbore or perforation wall and the well pressure is lower than the reservoir pressure so that a drawdown pressure exists for fluid to flow into the wellbore. Assuming that a perforation tunnel is a cylindrical hole sufficiently long so that it is not affected by the near-wellbore stresses, the maximum hoop stress at the wall of an open hole or a perforation tunnel can be calculated as:

$$\sigma_{\theta} = 3\sigma_1 - \sigma_2 - P_w + 2\eta(P_w - P_r) \quad (5)$$

Where σ_1 is the maximum far-field stress, σ_2 is the minimum far-field stress in the cross-section plane of the wellbore or perforation tunnel, P_r is the undisturbed reservoir pressure, and η is a poroelastic coefficient which is defined as:

$$\eta = \alpha \frac{1-2\nu}{2(1-\nu)} \quad (6)$$

where α is Biot's effective stress coefficient (Biot, 1941), ν is Poisson's ratio.

In an open hole completion, the most stable wellbore direction is found when the hoop stress at the borehole wall σ_{θ} as calculated in Eq. (5) reaches a minimum value. In a cased and perforated completion, the direction of the wellbore from which the most stable perforations originate is found when the hoop stress at the perforation wall reaches a minimum value. For an open hole, the optimum horizontal wellbore orientation in all stress regimes is the same as in the wellbore stability case. For a cased hole, the most stable orientation of a perforation depends on the maximum far-field stress and minimum far-field stress in the cross-section of the perforation tunnel. In a normal stress regime, if Eq. (7) is satisfied, the most stable perforations are vertical, and the well's azimuth does not impact the vertical perforation stability. Otherwise, the most stable perforations are horizontal perforations from a wellbore in the same direction as the maximum horizontal stress.

$$\sigma_V > \frac{1}{3}(4\sigma_H - \sigma_h) \quad (7)$$

In a reverse fault stress regime, if Eq. (8) is satisfied, the most stable perforations are vertical, and the well's azimuth does not impact the vertical perforation stability. If Eq. (8) is not satisfied, the most stable perforations are horizontal perforations from a wellbore parallel to the minimum horizontal stress.

$$\sigma_V > 4\sigma_h - 3\sigma_H \quad (8)$$

In a strike-slip stress regime, if Eq. (9) is satisfied, the most stable perforations are horizontal perforations from a wellbore in the same direction as the maximum horizontal stress. Otherwise, the most stable perforations are horizontal perforations from a wellbore aligned with the minimum horizontal stress.

$$\sigma_V > \frac{1}{4}(3\sigma_H + \sigma_h) \quad (9)$$

Horizontal Well Design for Optimum Hydraulic Fracturing Stimulation

In comparison with hydraulic fracturing of more porous and more permeable reservoir formations, where the main objective of the hydraulic fractures is to bypass the formation damage in the near-wellbore region,

hydraulic fracturing of tight reservoirs aims to provide more fracture surface area and therefore a larger effective reservoir drainage area. Since more fracture surface area is desirable in tight reservoirs and multiple transverse fractures provide larger fracture surface area compared to longitudinal fractures, horizontal wellbores in tight reservoirs should be placed in the minimum horizontal stress direction so that multiple transverse fractures can be created and propagated against the minimum in-situ stress.

Hydraulic fractures initiate from a wellbore when tensile stress at the wellbore wall exceeds the tensile strength of the rock. The initial tensile fractures are oriented in the direction normal to the tensile stress direction. Pressurizing a wellbore creates tensile stress in the tangential direction, and a longitudinal fracture is created when the tensile stress overcomes the tensile strength. Transverse fractures can also initiate from a wellbore if large enough axial tensile stress is created by pressurization of the pre-existing cracks, perforations, or notches (Murtadha et al., 2019). If a horizontal wellbore is oriented in the minimum horizontal stress direction and a longitudinal fracture initiates and propagates, it will turn from the axial direction to the transverse direction so that it can propagate against the minimum horizontal stress. When an open hole or a cased hole with a relatively long perforated interval is fractured, multiple fractures could be initiated from the wellbore and these fractures will compete for fluid when they propagate away from the wellbore. Fracture turning and multiple competing fractures create near-wellbore complexity that is detrimental to proppant transport and fracture conductivity.

In a cased hole oriented in the minimum horizontal stress direction, stresses generated at the wall of the perforation tunnel help to create transverse fractures. If a perforation tunnel is relatively long (greater than 3 times the wellbore diameter), the near-perforation stresses at the perforation tip are dominated by the far-field stresses, while at the perforation base, the near-perforation stresses are affected by the near-wellbore stresses. Under the most typical strike-slip stress regime in deep and tight reservoirs, the most likely direction for fractures initiated from the perforations is transverse (Waters and Weng, 1998). Limiting the perforated interval length helps to minimize the number of competing transverse fractures created in the perforated interval and therefore improves fracture conductivity.

When the reservoir formation is naturally fractured, placing the horizontal wellbore to cross as many open natural fractures as possible benefits fluid flow into the wellbore. However, interaction of hydraulic fracture and natural fractures creates complex fracture networks (Yew and Weng, 2015). Some hydraulic fractures may arrest the growth of a propagating hydraulic fracture while some others may allow hydraulic fractures to dilate and propagate the existing natural fractures so that they become part of the complex fracture network. The behavior of hydraulic fracture and natural fracture interaction depends on the direction and magnitudes of the in-situ stresses and the angle between a propagating hydraulic fracture and the intersecting natural fracture (Yew and Weng, 2015; Li et al., 2023). Complex fracture network simulation is required to determine the best wellbore orientation to achieve an optimum stimulated fracture network.

Integrated Approach

To determine the best horizontal well orientation considering all the requirements of wellbore stability, ROP, solids production and hydraulic fracturing in different phases of the wellbore construction and production, we developed an integrated approach to determine the optimum wellbore orientation requirements from considering each different objective and to recommend the final wellbore orientation considering all the requirements for different objectives. Fig. 1 shows the flow chart of our integrated approach to determining the optimum horizontal well orientation with considerations of all factors. The first step is to determine the most stable well orientation considering wellbore stability and solids production. The magnitudes of the in-situ stresses and the direction of the maximum horizontal stress are required input data for performing the decision process. With the stress input data and the equations shown in the sections above, the best horizontal well direction can be determined for wellbore stability and solids production considerations. If the planned completion type is open hole completion, the program provides the most stable wellbore direction

during both drilling and production, which are the same, assuming that the stress regime remains the same during drilling and production. The open hole stability during production could be different with reservoir pressure depletion if the stress regime changes after a certain amount of depletion. If cased and perforated completion is the selected completion type, the most stable perforation orientation and the required direction of the wellbore from which the perforations originate will be determined. The more stable wellbore direction during drilling and production could be different, depending on the stress regime and stress magnitudes. Similarly, reservoir pressure depletion will also complicate the choice of the most stable wellbore orientation if a stress regime change is induced during reservoir pressure depletion.

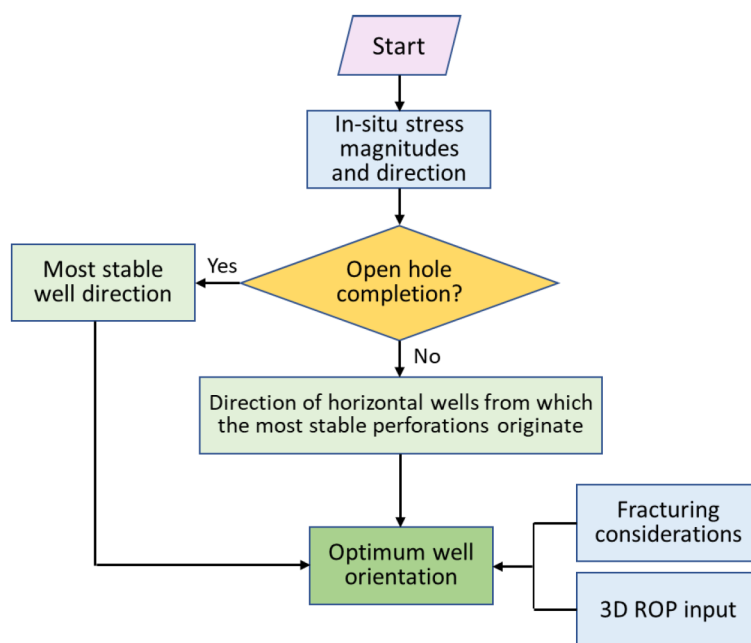


Figure 2—Flow chart to illustrate an integrated approach to determining the optimum horizontal well orientation considering risks of slow drilling, wellbore stability, solids production, and hydraulic fracturing.

If a horizontal well is hydraulically fractured, the optimum wellbore orientation is the minimum horizontal stress direction for creating multiple transverse fractures. A deviation of up to 30 degrees from the minimum horizontal stress direction will not significantly affect the creation of transverse fractures and the effectiveness of hydraulic fracturing stimulations (Jin and Shah, 2013; Kim and Abass, 1991; Lecampion et al., 2013; Wei and Economides, 2005). With the considerations of wellbore stability, solids production, and hydraulic fracturing, the optimum wellbore orientation can be determined, with some compromises between the different considerations if the required orientations are in conflict. The final step in the decision process is to insert the planned wellbore with the desired orientation into the 3D ROP model and find the final optimum wellbore orientation, also considering the ROP requirement. This may require the desired wellbore orientation determined from prior analysis to be adjusted, as long as the adjustment is not going to significantly compromise the other considerations, and if the ROP difference is large enough and a significant positive impact on drilling efficiency is expected.

Case Study

A tight clastic gas reservoir in the Middle East is developed with horizontal wells and multi-stage hydraulic fracturing stimulations. The reservoir consists mainly of sandstone with various amount of clay content and is characterized by high formation abrasiveness and high formation strength. UCS values of greater than 30,000 psi is frequently encountered in different parts of the reservoir. The most typical drilling challenges in the formation include slow drilling and fast bit wear which leads to low drill bit footage and frequent

bit changes. Wellbore instability problems are also encountered in some wells. The horizontal wells are either completed with open hole completions or cased and perforated completions. Sand production while producing natural gas occurs in some horizontal wells, regardless of completion types. The maximum horizontal stress direction is approximately N90°E in the reservoir. Horizontal wells are mostly drilled within a certain range of the minimum horizontal stress direction for creating transverse hydraulic fractures.

To address slow drilling and bit wear challenges, the 3D ROP prediction method was implemented in the reservoir, and the distribution of the predicted ROP and formation abrasiveness in the 3D model was created for the whole field. Horizontal wells planned to be drilled are designed considering the ROP and abrasiveness distribution in addition to other considerations. Fig. 3 shows the ROP distribution in part of the field where two horizontal wells W1 and W2 were planned. These two wells with close proximity to each other were planned to be drilled in the same reservoir zone and completed with casing and perforations. Both horizontal wells were planned to be drilled toward the North, the minimum horizontal stress direction, mainly to create transverse hydraulic fractures. We ran the horizontal well direction optimization program to determine the optimum well orientation from wellbore stability and solids production considerations. The 3D ROP distribution was subsequently used to identify potential improvements in drilling performance.

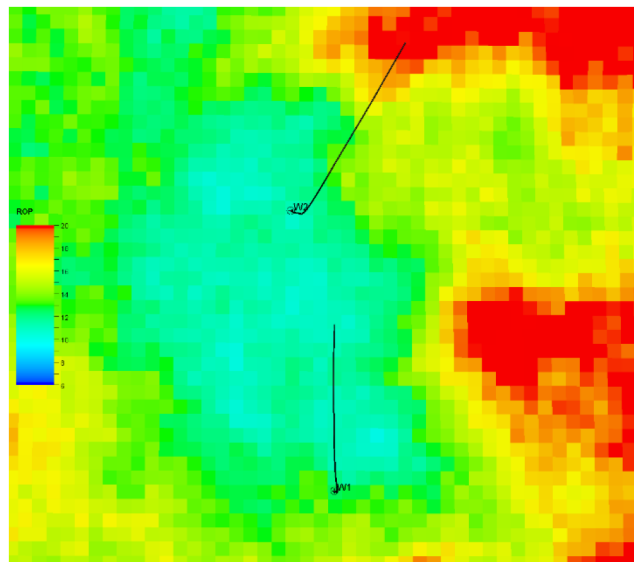


Figure 3—The predicted ROP distribution in the area with two horizontal wells planned.

The approximate magnitudes of in-situ stresses and the maximum horizontal stress azimuth in the reservoir zone at the location of these two wells are listed in Table 1. The stress magnitude data indicate that the stress regime is strike-slip. After running the optimum wellbore orientation program, the determined optimum well azimuth for drilling, solids production, and hydraulic fracturing are also listed in Table 1. Since both wells are cased holes, both solids production and hydraulic fracturing considerations require the well azimuth to be 0 or parallel to the minimum horizontal stress direction. The risk of wellbore instability is the lowest in the maximum horizontal stress direction. Considering that the requirement for hydraulic fracturing will affect the effectiveness of stimulation and therefore well production, it is more critical to place the wellbore in the preferred well azimuth of 0. The planned wellbores are thereafter compared with the 3D ROP distribution to evaluate potential drilling efficiency improvement by adjusting the well orientation. As can be seen from Fig. 3, the ROP distribution shows that the predicted ROP in the vicinity of W1 is relatively low, and there is no major ROP improvement expected if the wellbore azimuth is adjusted within 30 degrees. However, if W2 is adjusted slightly toward East from North, the ROP is expected to be improved significantly, especially in the second half of the lateral. In addition, orienting the wellbore toward the maximum horizontal stress direction will also improve wellbore stability while drilling. Considering

all the risks and potential improvement in ROP and wellbore stability, the well azimuth of W1 remained the same while the well azimuth of W2 was finalized to be 30 degrees from North, still within 30 degrees from the minimum horizontal stress direction, and therefore the impact on hydraulic fracturing should be very limited. With the final well trajectory plans for W1 and W2, a much-improved drilling performance is expected in W2.

Table 1—Input stress data at the two well locations and the result of optimum well orientation considering wellbore and perforation stability

Input Data	
Maximum horizontal stress (psi)	17000
Minimum Horizontal Stress (psi)	11000
Vertical stress (psi)	13000
Maximum horizontal stress azimuth (degree)	90
Optimum Well Orientation Result	
Optimum well azimuth during drilling (degree)	90
Optimum well azimuth during production – open hole (degree)	90
Optimum well azimuth during production – cased hole (degree)	0
Optimum perforation direction during production	Horizontal
Optimum well azimuth for hydraulic fracturing	0

After these two wells were drilled, the drilling performance of both wells was evaluated. Table 2 shows the drilling performance comparison between these two wells. As shown in the data, five drill bits were used in drilling the W1 lateral while only two were required for drilling W2. All other data, including total footage, number of days, footage per day, and footage per run, clearly show that the drilling efficiency of W2 is significantly higher than that of W1.

Table 2—Drilling performance comparison of two horizontal wells drilled in a tight reservoir

Well	Number of Drill Bits	Total Footage (ft)	Number of Days	Footage (ft) /Day	Footage (ft) /Run
W1	5	2148	16.4	131	429.6
W2	2	3291	15.8	207.8	1645.5

Ignoring all other parameters affecting wellbore stability, if these two wells are compared just from the well azimuth point of view, W2 should be more stable than W1 because W1 is in the least favorable direction. However, since the stress magnitudes and rock strength around these two wellbores may not be the same, it is not straightforward to compare the wellbore stability conditions of these two wells. Fig. 4 shows the bit size (BS) and caliper logs (Caliper) of both laterals. It shows a similar wellbore size compared to the bit size in most parts of the laterals. However, the first 300 ft of W1 is significantly more enlarged, which should have been partially contributed to by the least favorable wellbore orientation in terms of wellbore stability.

There is no data to compare hydraulic fracture geometry or solids production between these two wells yet, just from ROP and wellbore stability perspectives, the integrated approach for horizontal well direction optimization has shown great value in improving drilling performance without compromising other objectives of the wells.

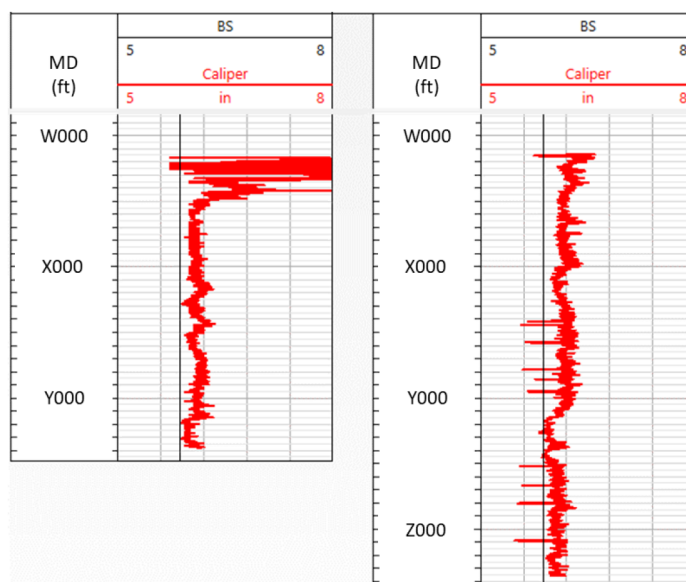


Figure 4—Comparison of caliper logs between W1 (left) and W2 (right). The first 300 ft of the lateral in W1 shows more significant wellbore enlargement, while the rest of the laterals in both wells have similar hole sizes. BS is the bit size in the wellbore section.

Conclusions

Multiple risks exist in different phases of horizontal well drilling, completion, and production in tight reservoirs. The direction and magnitudes of in-situ stresses affect the stability of wellbores and perforation tunnels during drilling and production. Horizontal well orientation significantly affects hydraulic fracture geometry and stimulation effectiveness. Depending on the reservoir permeability and the desired fracture geometry, horizontal wells need to be placed either parallel to the maximum horizontal stress or normal to it. The ROP and drilling efficiency of horizontal wells drilled in a tight reservoir can be improved using a 3D model of predicted ROP and formation abrasiveness.

An integrated method has been developed to determine the optimum wellbore orientation considering all the risks of wellbore stability, solids production, hydraulic fracture geometry, and drilling performance. With the input data of in-situ stresses, a horizontal well orientation optimization program determines the wellbore orientation with the most stable wellbore during drilling and open hole or perforation tunnels during production. With the additional considerations of hydraulic fracture geometry and drilling performance, the final optimum horizontal well orientation can be determined. The integrated approach has been validated with a case study where two horizontal wells were planned in a tight reservoir. Data from both wells after drilling validated the approach and proved that the approach could bring significant value to horizontal well planning.

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